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MATH.* Statistics:-

Significance level (α) — probability of rejecting the NULL hypothesis (H_0) when it is actually True.
 — It represents a risk of making Type I error.

Power of a test

↓ probability that the test rejects the NULL Hypothesis.

Two-tailed Test

two tailed $\mu = \mu_0$

$$H_0: \mu = \mu_0$$

$$H_1: \mu \neq \mu_0$$

one tailed when any one of the Tests are accepted eg: $\mu > \mu_0$
 $\mu < \mu_0$

* Confidence Interval — considers only mean/proportion.

One sample Z-Test:-

$$Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$$

$\bar{X}$ → mean

μ_0 → population mean

σ → std. deviation

n → total no of population or (count)

$H_1: \mu \neq 100$ (depicts one tailed)

Rejection of null hypothesis — when t is lesser than the calculated value.

Left tailed $Z_{cal} < Z_{tab} \rightarrow \text{Reject } H_0.$

If 2 tailed consider $\Rightarrow |Z_{cal}|.$

Two sample Z Test:

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

σ_1, σ_2 , is the population standard deviation

Hypothesis is rejected when, t is lesser than the calculated value.

Two tailed = $\pm$ value

left tailed (single tailed) = +value.

right tailed (single tailed) = -value.

Problem 1:

A company manufactures resistors with a claimed mean resistance of 100 Ω

Population std deviation $\sigma = 2.5 \Omega$

Random sample, $n = 60$, Sample mean $\bar{X} = 99.4 \Omega$

Soln:- Given

$$\sigma = 2.5 \Omega$$

$$n = 60$$

$$\alpha = 0.01$$

$$\bar{x} = 99.4 \Omega$$

$$\mu_0 = 100 \Omega$$

(i) Check if True mean differs from 100Ω .

$$Z_{\text{test}} = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

$$\Rightarrow \frac{99.4 - 100}{2.5 / \sqrt{60}} \Rightarrow \frac{0.6}{2.5 / \sqrt{6 \times 10}}$$

$$\Rightarrow \frac{0.6}{2.5 / 2 \times \sqrt{15}}$$

$$\Rightarrow \left(\frac{0.6 \times 2 \times \sqrt{15}}{2.5} \right)$$

$$Z_{\text{cal}} \Rightarrow -1.86 \quad (\text{Calculated value of } Z_{\text{test}})$$

$Z_{\text{tabulated}}$: for

$$2 \text{ tails} \Rightarrow \alpha/2 \quad \alpha \rightarrow 0.01$$

$$\text{then } \alpha/2 = 0.005$$

$\alpha \rightarrow$ determines whether single or 2 tailed.

$$Z_{\text{tab}} = 2.58$$

$$|Z_{cal}| < Z_{tab}$$

i.e. $1.86 < 2.58$.

Hence the hypothesis is failed to be rejected

i.e. the hypothesis is accepted

Hence, the NULL Hypothesis is accepted.

Hence True mean differs from 100.2

(P_i)

$$Z\text{-test} \Rightarrow \bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$$Z_{test} = \frac{99.4 + 2.58 \times 2.5}{\sqrt{60}}$$

$$\text{When } \frac{99.4 + 2.58 \times 2.5}{\sqrt{60}} = 98.57$$

$$99.4 - 2.58 \times \frac{2.5}{\sqrt{60}} = 100.23$$

Hence the interval is $(98.57, 100.23)$

Soln: Mean lies in the above region, i.e. $(98.57, 100.23)$, hence it cannot be rejected.
 applied + and - values separately.

applicable
for
2 tail
and
1 tail

$$|Z_{cal}| < |Z_{tab}| \Rightarrow \text{Null hypothesis is accepted}$$

$$|Z_{cal}| < |Z_{tab}|$$

Right tail & 2 tail

Left tail $\rightarrow$ +ve
-ve.

(iii) Interpret ~~the~~ result
mean $\rightarrow 100\%$

$$\mu_0 = 250 \text{ ml}$$

$$n = 36 \text{ cups.}$$

$$\sigma = 5.$$

$$\bar{X} = 247 \text{ ml}$$

$$H_0 \Rightarrow \mu = 250 \text{ ml}$$

$$\mu \neq 250 \text{ ml.}$$

$$\begin{array}{r} 3.6 \\ 5 \overline{) 18} \\ \underline{-15} \\ 30 \\ \underline{-30} \\ 0 \end{array}$$

$$Z = \frac{247 - 250}{5/\sqrt{36}} \Rightarrow \frac{-3}{5/6} \Rightarrow$$

$$= \frac{-3 \times 6}{5} = \frac{-18}{5} = -3.6$$

$$Z_{\text{cal}} = -3.6.$$

$$Z_{\text{tab}} \text{ at } \alpha = 0.05$$

$$\Rightarrow \cancel{2.40}$$

$$Z_{\alpha/2} = 1.96.$$

Calculated value $>$ Tabulated value of
 Z thus, the hypothesis is rejected.

(*) 95% confidence.

$$CI = \bar{X} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$$\Rightarrow 247 \pm \frac{1.96 \times 5}{6}$$

$$\Rightarrow (247.87, 248.63)$$

250 ml is not in the range of the interval (247.87, 248.63) and hence the hypothesis proposed is rejected.

2 Sample Z-test :-

Battery A and B

$$\sigma_A = 3.5 \text{ hours.}$$

$$\sigma_B = 4.0 \text{ hours.}$$

Q1 - says $\Rightarrow$ mean life time

Null Hypothesis : if ~~the~~ Mean life time of battery of brand A, μ_1 and Mean lifetime of battery of brand B, μ_2 .

$$CI = \bar{X} \pm Z_{\alpha} \left(\sigma / \sqrt{n} \right).$$

$$Z_{\text{calculated}} = \frac{41.8 - 39.6}{\sqrt{\frac{3.5^2}{40} + \frac{4^2}{50}}} \Rightarrow \underline{\underline{2.86}}$$

$$Z_{\alpha} \text{ value.} = \frac{Z_{0.01}}{2} = \underline{\underline{2.58}}$$

(Z_{calculated} with α value).

$Z_{\text{cal}} > Z_{\text{tab}}$, Hence the Null hypothesis, H_0 is rejected, Hence their battery lifetime differ.

P.T.O.

Z-test

2tailed :-

$$(\bar{x}_1 - \bar{x}_2) \pm Z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

for 2 tailed, we consider.

2 values of mean population
along with 2 different values of
standard deviation and n value,
count of the population.

$$\bar{x}_1 = 41.8 \quad \bar{x}_2 = 39.6 \quad Z_{\alpha/2} = 2.58$$

$$Z_{tab} = \sigma_1 = 3.5 \quad \sigma_2 = 4.0 \quad n_1 = 40 \quad n_2 = 50$$

$$CI \quad Z_{tab} = (41.8 - 39.6) \pm 1.96 \sqrt{\frac{(3.5)^2}{40} + \frac{(4.0)^2}{50}}$$

$$\Rightarrow 2.2 \pm 1.96 \times 0.79$$

$$\Rightarrow 2.2 + (1.96 \times 0.79)$$

Q1:- Ho-

or

$$2.2 - (1.96 \times 0.79)$$

$$\text{result} = (0.649, 3.751)$$

range of CI would hence
be $\Rightarrow (0.649, 3.751)$

↓
result

$$\text{Approx } (0.65, 3.75)$$

Brand A lasts significantly in Brand B.

Q. A simple random sampling survey in respect of monthly earnings of semi-skilled workers in two cities gives the following statistical information.

City	Mean Monthly earnings	SD of sample data of monthly earnings	Size of sample
A	695 Rs.	40	200
B	710 Rs.	60	175

Test the hypothesis at 5% significance level that there is no difference b/w monthly earnings of workers of 2 cities.

Soln

$$\bar{X}_A = 695$$

$$\bar{X}_B = 710$$

$$\sigma_A = 40$$

$$\sigma_B = 60$$

$$n_A = 200$$

$$n_B = 175$$

Hypothesis

$$H_0 = \mu_A = \mu_B$$

$$H_1 = \mu_A \neq \mu_B$$

(2-tailed)

$$\frac{610}{710}$$

$$-695$$

$$15$$

$$Z = \frac{\bar{X}_A - \bar{X}_B}{\sqrt{\frac{\sigma_A^2}{n_A} + \frac{\sigma_B^2}{n_B}}} = \frac{695 - 710}{\sqrt{\frac{40^2}{200} + \frac{60^2}{175}}}$$

$$\Rightarrow -15$$

$$= -2.809$$

$$\sqrt{\frac{1600}{200} + \frac{3600}{175}}$$

$$\frac{Z_{\alpha}}{2} = \frac{20.05}{2} = 1.96$$

$$|Z_{cal}| > Z_{tab}$$

Hence reject H_0 .

$$|Z_{cal}| > Z_{tab} \rightarrow \text{reject } H_0$$

* Therefore, there is a difference b/w monthly earnings of 2 cities. (Since Null hypothesis is rejected)

Qn).

Null hypothesis $\Rightarrow$ at least 1200, $\Rightarrow = 1200$

Alternate hypothesis $\Rightarrow < 1200$.

$$n = 40$$

$$\bar{X} = 1185$$

$$\sigma = 30$$

$$\mu_0 = 1200$$

$$1200$$

$$-1185$$

$$15$$

$$H_0 \geq 1200$$

$\rightarrow$ Null hypothesis

$$H_1 < 1200$$

$\rightarrow$ Alternate hypothesis

$$Z = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}} = \frac{1185 - 1200}{\frac{30}{\sqrt{40}}}$$

$$\Rightarrow \frac{-15}{\frac{30}{\sqrt{40}}} = - \frac{15 \times \sqrt{40}}{30}$$

$$Z_{test} = -\sqrt{10}$$

$$Z_{tab} = Z_{\alpha} = Z_{0.05} = \underline{-1.645}$$

$$|Z_{cal}| = 3.65 > 1.645 = |Z_{tab}|$$

Hence, H_0 is rejected.

Hence, Mean life is less than 1200.

②. Determine whether factory's claim is supported by the sample.

Q

T-test

- Used to compare sample mean.

Types of t-test

- One sample t-test
- Two sample t-test

$$H_0: \mu = \mu_0$$

$$H_1: \mu \neq \mu_0$$

* One sample t-test	$t = \frac{\bar{X} - \mu_0}{s / \sqrt{n}}$
------------------------	--

Compare $|t|$ to t_{n-1} and $\alpha/2$ by Ttable

In one sample degree = -t-test

$$t_{n-1} \alpha/2$$

* 2 sample t-tests.

↳ Check if population variances are equal or not.

* If variances are equal:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

Pooled standard deviation:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$\text{Degrees of freedom} = df = n_1 + n_2 - 2$$

* Unequal variances

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Degree of freedom :- Welch - Satterthwaite approximation

$$df \approx \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$$

One tail & 2 tail considered in t-test.

Qn) A process scheduler claims mean CPU utilization = 70%. Sample of 10 users: 68, 72, 69, 70, 65, 71, 68, 74, 69, 67. At $\alpha = 0.05$, verify claim.

Soln :- Null hypothesis H_0 :- Mean CPU utilization = 70%

Alternate hypothesis H_1 :- Mean CPU utilization $\neq 70\%$

$$\bar{X} = \frac{68 + 72 + 69 + 70 + 65 + 71 + 68 + 74 + 69 + 67}{10}$$

$$\bar{X} = 69.3$$

Standard deviation :- $\sqrt{\frac{\sum (X - \bar{X})^2}{n-1}}$

$$\begin{array}{r} 29 \times 29 \\ 82 \times 22 \\ \hline 189 \\ 540 \\ \hline 729 \end{array}$$

$$\begin{aligned} & (68 - 69.3)^2 + (72 - 69.3)^2 + (69 - 69.3)^2 + \\ & (70 - 69.3)^2 + (65 - 69.3)^2 + (71 - 69.3)^2 + \\ & (68 - 69.3)^2 + (74 - 69.3)^2 + (69 - 69.3)^2 \\ & + (67 - 69.3)^2 \end{aligned}$$

9.

$$\begin{array}{r} 26.03 \\ 3 \overline{) 60.1} \\ \underline{-6} \\ 0010 \\ \underline{-3} \end{array}$$

~~$(3+23+0.3+0.7+4.3+4.7)$~~

$$\begin{aligned} & \sum 1.69 + 7.29 + 0.09 + 0.49 + 18.49 + 289 + \\ & 1.69 + 22.09 + 0.09 + 5.29. \end{aligned}$$

9

$$\Rightarrow \sqrt{\frac{1 \times (60.1)}{9}} \Rightarrow \underline{\underline{2.58}}$$

$\Rightarrow \text{soln} = \underline{\underline{2.58}}$

$$\text{thus } T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} = \frac{69.3 - 70}{2.58/\sqrt{10}} = 0.857$$

$t_{cal} =$

degree of freedom is ~~$n-1=10$~~

$\Rightarrow 10$

just for one tail
 $n-1 = 9$

$= 1.8$ (one tail)

$t_{tabulated} = t_{9, \frac{0.05}{2}} = t_{9, 0.025} =$ (one tail problem)

Hence $t_{cal} = -0.858$.

$t_{tab} = 2.262$.

$$(|-0.858|) < t_{tab} (2.262)$$

Hence Failed to Reject H_0 .

Hence H_0 is accepted. Mean CPU Life time is 70% (accepted)

MATHT-test continuationIndependent Samples t-test: (Model Accuracy)

A researcher evaluates 2 ML models using 10-fold cross-validation:

Model A: 91 89 90 92 88 93 90 91 89 90

Model B: 88 85 87 86 88 89 86 87 85 86

Test whether there is significant difference in mean accuracies at $\alpha = 0.05$. Assume the two samples are independent & variances are equal.

	X_A	$X_A - \bar{X}_A$	X_B	$X_B - \bar{X}_B$	$(X_A - \bar{X}_A)^2$	$(X_B - \bar{X}_B)^2$
1	91	0.7	88	1.3	0.49	1.69
2	89	-1.3	85	-1.7	1.69	2.89
3	90	-0.3	87	0.3	0.09	0.09
4	92	1.7	86	-0.7	2.89	0.49
5	88	-2.3	88	1.3	5.29	1.69
6	93	2.7	89	2.3	7.29	5.29
7	90	-0.3	86	-0.7	0.09	0.49
8	91	0.7	87	0.3	0.49	0.09
9	89	-1.3	85	-1.7	1.69	2.89
10	90	-0.3	86	-0.7	0.09	0.49
					<u>20.10</u>	<u>16.1</u>

$$\bar{X}_A = 90.3$$

$$\bar{X}_B = 86.7$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{sp \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$S_A^2 = \frac{\sum (X_A - \bar{X}_A)^2}{n_A - 1} \Rightarrow \frac{20 \cdot 10 = 2233}{9} = 2.2333 \quad \begin{matrix} n_A = 10 \\ n_A - 1 = 9 \end{matrix}$$

$$S_B^2 = \frac{\sum (X_B - \bar{X}_B)^2}{n_B - 1} \Rightarrow \frac{16 \cdot 1}{9} \Rightarrow \underline{1.789}$$

Thus $S_A^2 = 2.2333$ $S_B^2 = 1.789$

$$t = \frac{\bar{X}_A - \bar{X}_B}{sp \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \Rightarrow \frac{90.3 - 86.7}{sp \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$sp = \sqrt{\frac{(n_A - 1)S_A^2 + (n_B - 1)S_B^2}{n_A + n_B - 2}}$$

$$\begin{aligned} & \frac{20.0997}{16.1010} \Rightarrow \frac{(9 \times 2.2333) + (9 \times 1.789)}{18} \Rightarrow \frac{20.0997 + 16.101}{18} \Rightarrow \frac{36.2007}{8} \end{aligned}$$

$$t = \frac{90.3 - 86.7}{\sqrt{14.528} \sqrt{\frac{1}{10} + \frac{1}{10}}} \Rightarrow \underline{\underline{1.417}}$$

degree of freedom $df = n_A + n_B - 2$
 $\Rightarrow 10 + 10 - 2$
 $= 18.$

$$t_{18, 0.05} = 2.101.$$

$$t_{cal} = 5.689$$

$$t_{cal} = 5.689 > t_{tab} = 2.101.$$

Hence $\Rightarrow$ there is significance diff.
 i.e. the Hypothesis H_0 is rejected.

Qn-2) Equal variances {Pooled t test}

Qn) A group of seven chickens reared on a high protein diet of 12, 15, 11, 16, 14, 4 and 16 ounces; a second group of five chickens of same age similarly treated except that they receive a low protein diet of 8, 10, 14, 10 and 13 ounces. Test at 5 per cent level whether there is significant evidence that additional protein has increased the weight of the chickens.

According to the above question - there are 2 hypotheses

Hypothesis

$H_0 \rightarrow$ weights of high protein and low protein diet chickens are same.

$H_1 \rightarrow$ alternate hypothesis: Weight of high protein diet chickens are weight of low protein diet chickens.

$$\text{degree of freedom} = n_1 + n_2 - 2$$

$$\Rightarrow 7 + 5 - 2 = 12 - 2 = 10.$$

5% Significance level.

$$t_{10, 0.05} = \underline{\underline{1.812}} \quad (t_{\text{tabulated}})$$

$$\text{Similarly} \Rightarrow t_{\text{calculated}} \Rightarrow \frac{\bar{x}_1 - \bar{x}_2}{s_1^2}$$

$$\frac{s_1^2}{6} = \frac{22}{6} = 3.67.$$

x_1	$x_1 - \bar{x}_1$	$(x_1 - \bar{x}_1)^2$	x_2	$(x_2 - \bar{x}_2)$	$(x_2 - \bar{x}_2)^2$
12	-2	4	8	-3	9
15	1	1	10	-1	1
11	-3	9	14	3	9
16	2	4	10	-1	1
14	0	0	13	2	4
14	0	0			
16	2	4			
$\bar{x}_1 = 14$		<u>22</u>	$\bar{x}_2 = 11$		

$$s_1^2 = \frac{22}{7} = 3.67$$

$$s_2^2 = \frac{24}{4} = 6$$

Hence,
$$s_p^2 = \frac{6 \cdot (3.67) + 4(6)}{10}$$

$$s_p^2 = 4.6$$

$$s_p = \sqrt{4.6}$$

$$t = \frac{14 - 11}{\sqrt{4.6} \sqrt{\frac{1}{7} + \frac{1}{4}}} \Rightarrow \underline{\underline{2.389}}$$

Hence $t_{cal} 2.389 > t_{tab} = 1.812$

Hence, H_0 is rejected, i.e. $H_1 \rightarrow$ true.

Thus Weight of high protein diet chickens are more than weight of low protein diet chickens.